

Element L: Presentation of Designer's Recommendations

L1. Content of Recommendations

Recommendation 1: Redesign the Mouthpiece Interface

Taking into account the challenges we faced with the mouthpiece design, we believe implementing a modular mouthpiece system with interchangeable sizes would improve user comfort and measurement accuracy because it would accommodate different user anatomies while maintaining proper sealing. Our testing revealed inconsistent readings when different users operated the device, indicating that a one-size-fits-all approach compromises measurement reliability.

Recommendation 2: Enhance Airtight Sealing Mechanisms

Taking into account the sealing issues encountered during prototype testing, we recommend incorporating silicone gaskets at all connection points would improve measurement accuracy because it would eliminate air leakage that currently affects

pressure readings by up to ± 2 inHg. Expert feedback specifically highlighted that our current connection between the gauge and tubing is a weak point for potential leakage.

Recommendation 3: Strengthen the 3D-Printed Casing Design

Taking into account our durability testing results, we recommend reinforcing stress points in the 3D-printed housing with additional material thickness would improve the device's longevity because our current prototype showed signs of strain after repeated use.

Specifically, the connection point between the gauge mount and main body would benefit from additional material support.

Recommendation 4: Implement Quality Control Standards

Taking into account the variability between prototype units, we recommend establishing a standardized quality control process with measurement calibration checks would improve consistency across multiple devices because our classroom implementation requires reproducible results across 10-20 units. Currently, each device requires individual calibration adjustments.

Recommendation 5: Incorporate User-Friendly Design Elements

Taking into account stakeholder feedback from instructors, we recommend adding visual guidelines and clear measurement markers would improve the educational value of the device because it would help students better understand target respiratory values. Expert

feedback suggested that visual cues for "normal" values would enhance the learning experience.

L2. Detailed Plans for Implementation of Improvements

Implementation Plan for Modular Mouthpiece System

1. Design three standard mouthpiece sizes (small, medium, large) with identical connection interfaces
2. Utilize medical-grade silicone for the mouthpiece material to ensure comfort and easy sanitization
3. Create a bayonet-style locking mechanism to ensure secure attachment while allowing for quick disconnection
4. Include clear size markings on each mouthpiece to facilitate selection
5. Test each size with at least 10 different users to validate comfort and sealing effectiveness
6. Incorporate disposable filters between the mouthpiece and measurement tube to address hygiene concerns
7. Package each device with all three sizes to ensure classroom versatility

Implementation Plan for Enhanced Sealing Mechanisms

1. Replace current pressure gauge connection with a threaded NPT fitting that accommodates silicone gaskets
2. Install O-rings at each connection point, specifically sized to match component dimensions
3. Apply food-grade silicone sealant to reinforce permanent connections
4. Create a standardized leak testing procedure using a calibrated pressure source
5. Implement a water submersion test protocol to visually identify any air leaks
6. Design a specialized tool for proper tightening of connections to prevent over-tightening damage
7. Document optimal torque specifications for each connection point to ensure consistent assembly

Implementation Plan for Strengthened Casing Design

1. Increase wall thickness from 2mm to 3.5mm at identified stress points
2. Add 45-degree fillets to all interior corners to distribute stress more effectively
3. Implement a ribbed support structure inside the casing to enhance structural integrity
4. Adjust 3D printing parameters to 35% infill density (up from current 20%)
5. Change print orientation to optimize layer alignment with primary stress vectors

6. Revise the gauge mounting mechanism to distribute force over a larger surface area
7. Conduct accelerated lifespan testing by simulating 5 years of typical classroom use

Implementation Plan for Quality Control Standards

1. Design and 3D print a calibration jig that connects to a verified pressure source
2. Create a digital documentation system to track each device's calibration history
3. Establish a three-point calibration procedure (zero, mid-range, maximum pressure)
4. Implement color-coded calibration stickers to indicate when recalibration is needed
5. Develop a troubleshooting flowchart for identifying common calibration issues
6. Create a detailed assembly checklist to ensure consistency in manufacturing
7. Establish maximum acceptable variance thresholds (± 0.5 inHg) for quality assurance

Implementation Plan for User-Friendly Design Elements

1. Integrate color-coded measurement zones on the gauge face (low/normal/high)
2. Design and apply a clear instructional label with step-by-step usage guidelines
3. Create a QR code that links to video demonstrations of proper technique
4. Add tactile indicators to guide proper hand placement during operation

5. Develop accompanying educational materials explaining respiratory measurement significance
6. Design a compact quick-reference guide to attach to the storage case
7. Create a digital companion app for recording and analyzing measurement data over time